

CSE-251

ELECTRONIC CIRCUITS

LECTURE-07

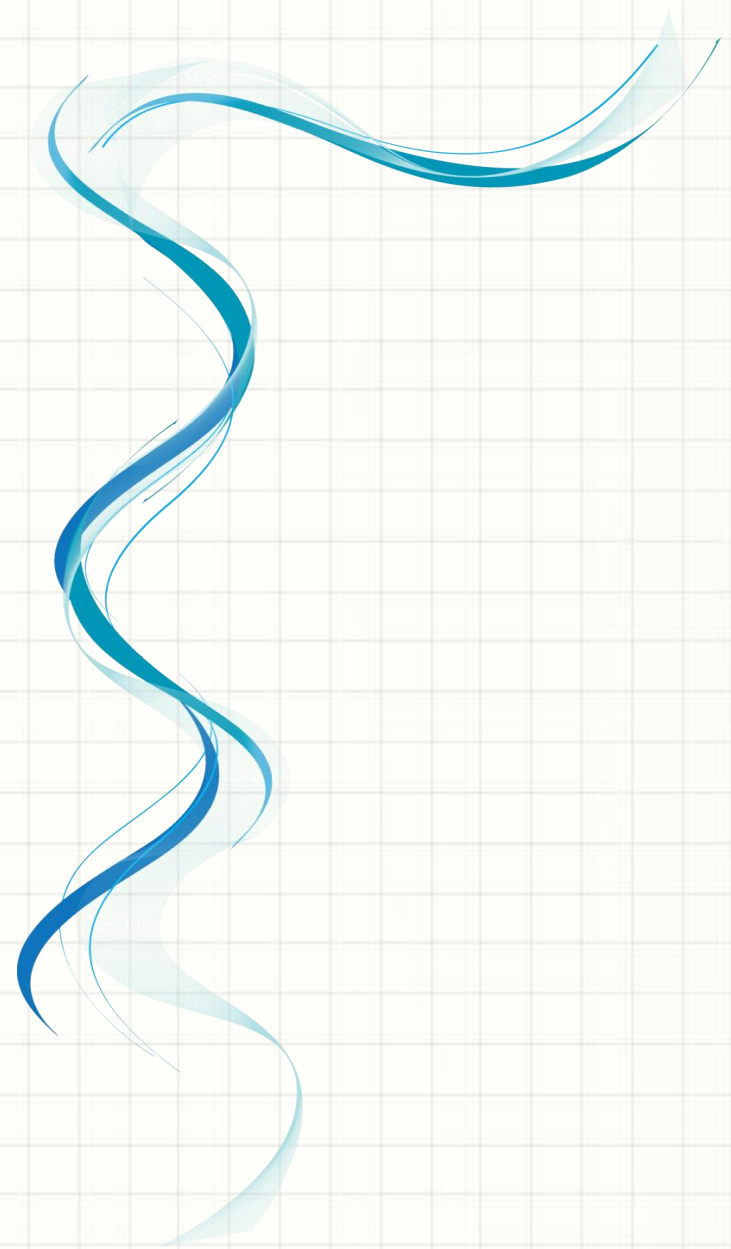
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Today's Overview:

- ❑ *Diode as a Rectifier*
- ❑ *Ripple*



Diode as a Rectifier



Rectification:

The main application of p-n junction diode is in rectification circuits. These circuits are used to describe the conversion of ac signals to dc in power supplies.

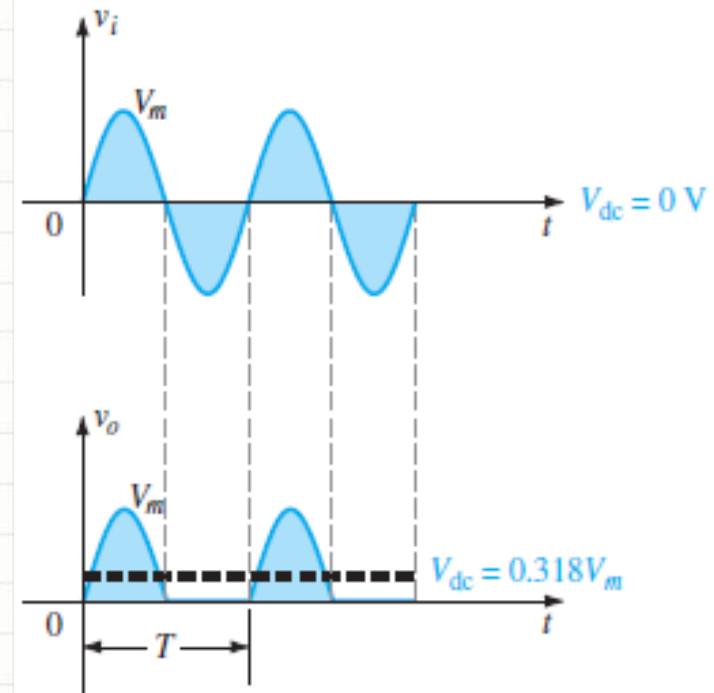
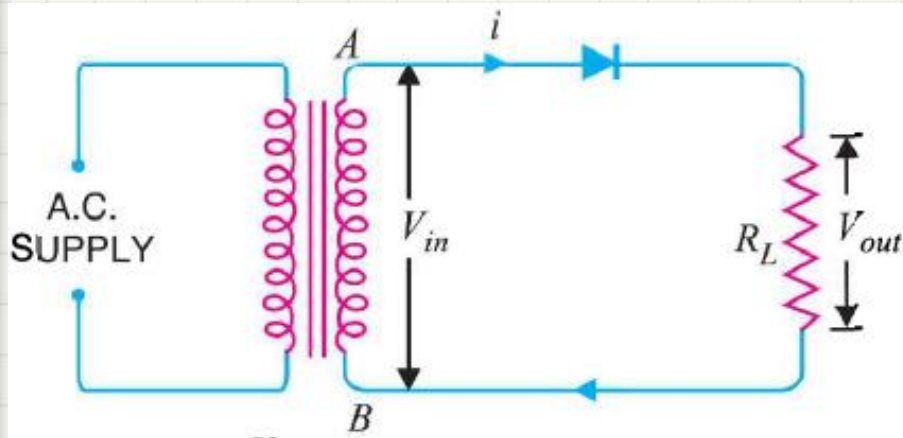
There are two primary methods of diode rectification:

- Half Wave Rectifier
- Full Wave Rectifier

Half-wave Rectifier:

The **Half-wave rectifier** has both positive and negative cycles. In a Half-wave rectifier, one half of each ac input cycle is rectified. During the positive half of the input, the current will flow from positive to negative which will generate only a positive half cycle of the ac supply. Here the diode gives little resistance.

In the second half cycle, the diode will be reverse biased and it provides high resistance. Thus, at the output side, there will be no current generated, and we cannot get power at the load resistance.



Half-wave Rectifier:

Advantages of Half Wave Rectifier:

- Affordable
- Simple connections
- Easy to use as the connections are simple
- Number of components used are less

Disadvantages of Half Wave Rectifier:

- Ripple production is more
- Utilization of the transformer is very low
- The efficiency of rectification is low

Half-wave Rectifier:

Applications of Half Wave Rectifier:

- ***Power rectification:*** Half wave rectifier is used along with a transformer for power rectification as powering equipment.
- ***Signal demodulation:*** Half wave rectifiers are used for demodulating the AM signals.
- ***Signal peak detector:*** Half wave rectifier is used for detecting the peak of the incoming waveform.

Half-wave Rectifier:

Average Value for full time period:

$$V = V_m \sin \omega t$$
$$\bar{V} = V_{av} = \frac{V_m}{\pi} = 0.318 V_m$$

Half-wave Rectifier:

RMS Value for full time period:

$$V = V_m \sin \omega t$$
$$V_{rms} = \frac{V_m}{2} = 0.5 V_m$$

Half-wave Rectifier:

Form Factor:

Form factor (F.F) is the ratio between RMS value and average value, as shown in the formula below:

$$F.F = \frac{RMS\ value}{Average\ value}$$

The form factor of a half wave rectifier is equal to 1.57 (i.e. F.F= 1.57).

Half-wave Rectifier:

Efficiency:

Rectifier efficiency (η) is the ratio between the output DC power and the input AC power.

The formula for the efficiency is equal to:

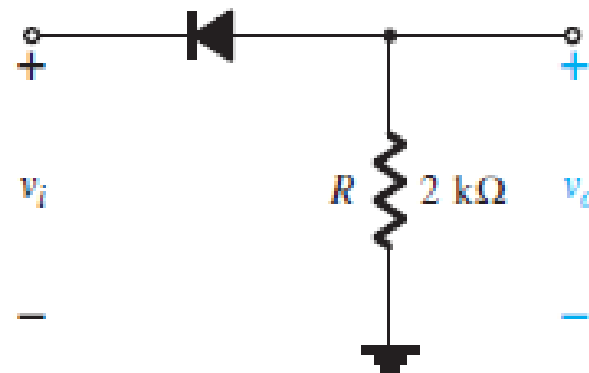
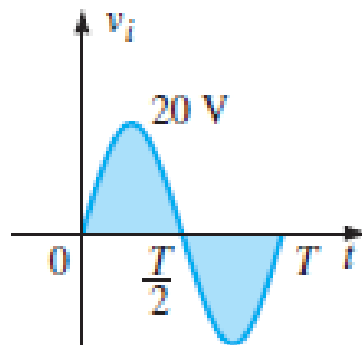
$$\eta = \frac{P_{dc}}{P_{ac}}$$

The efficiency of a half wave rectifier is equal to 40.6% (i.e. $\eta_{\max} = 40.6\%$)

Problem:

EXAMPLE 2.16

- Sketch the output v_o and determine the dc level of the output for the network of Fig. 2.49.
- Repeat part (a) if the ideal diode is replaced by a silicon diode.



Problem:

Solution:

- a. In this situation the diode will conduct during the negative part of the input as shown in Fig. 2.50, and v_o will appear as shown in the same figure. For the full period, the dc level is

$$V_{dc} = -0.318V_m = -0.318(20 \text{ V}) = -6.36 \text{ V}$$

The negative sign indicates that the polarity of the output is opposite to the defined polarity of Fig. 2.49.

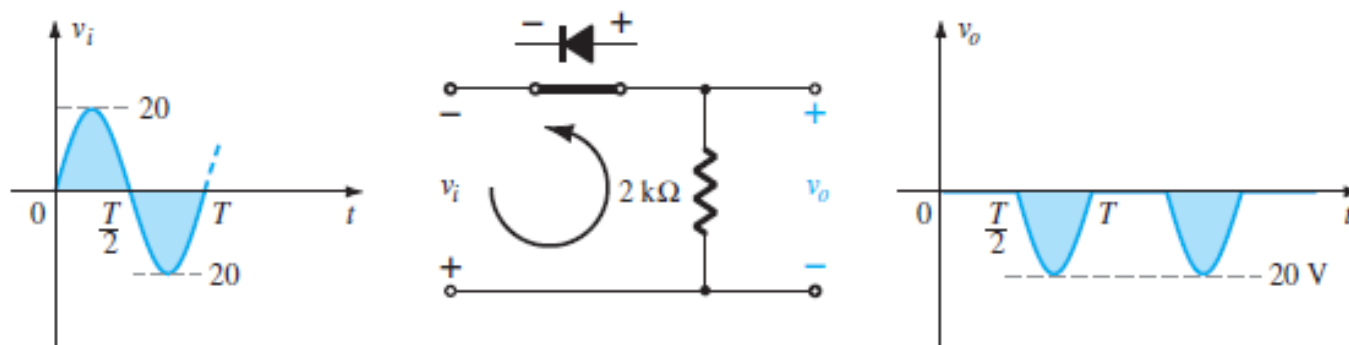
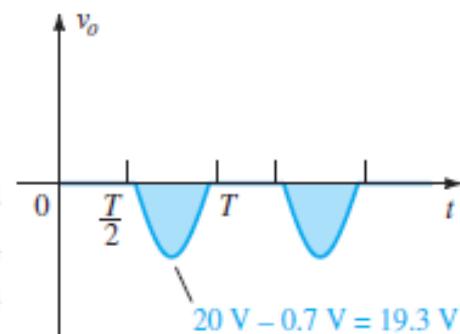


FIG. 2.50

Resulting v_o for the circuit of Example 2.16.

- b. For a silicon diode, the output has the appearance of Fig. 2.51, and

$$V_{dc} \cong -0.318(V_m - 0.7 \text{ V}) = -0.318(19.3 \text{ V}) \cong -6.14 \text{ V}$$



Full-wave Rectifier:

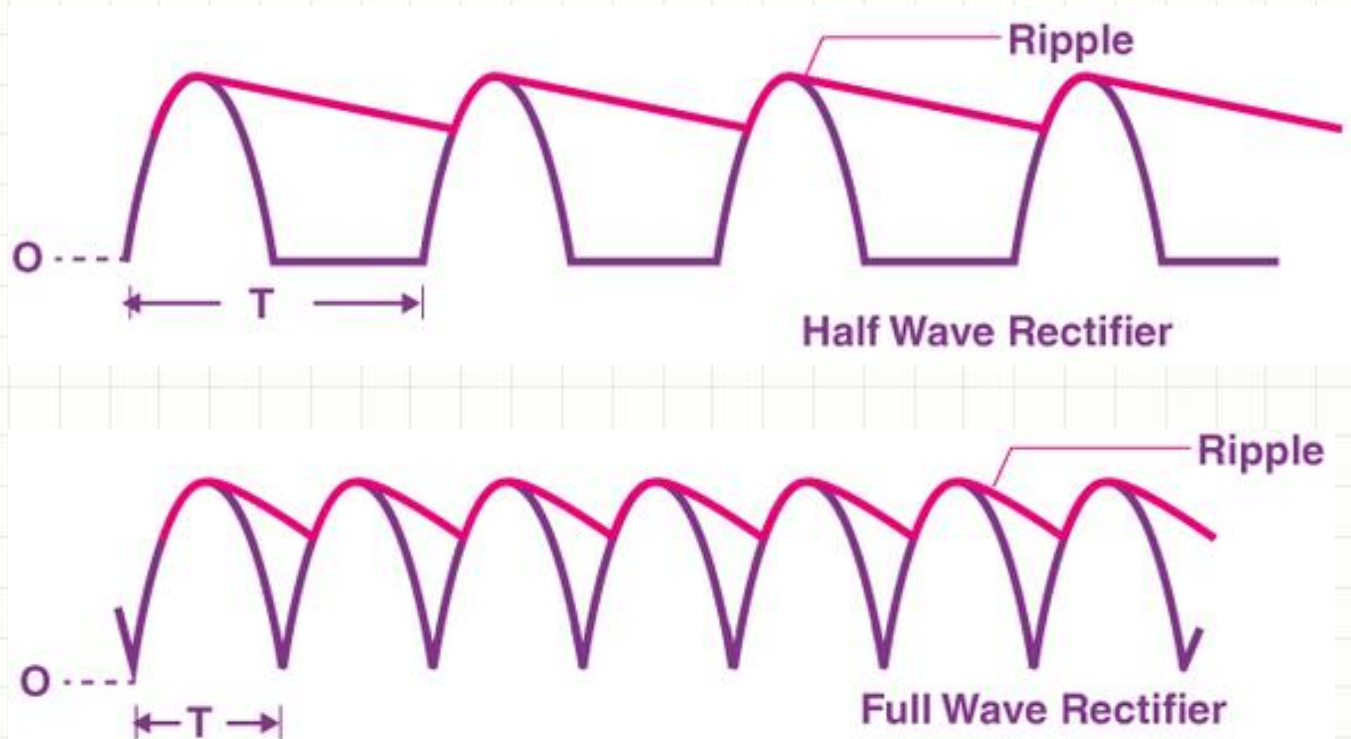
Full-wave rectifier circuits are used for producing an output voltage or output current which is purely DC.

The main advantage of a full-wave rectifier over half-wave rectifier is that the average output voltage is higher in full-wave rectifier, there is less ripple produced in full-wave rectifier when compared to the half-wave rectifier.

Ripple:

A **Ripple** is defined as the fluctuating AC component in the rectified DC output.

The rectified DC output could be either DC current or DC voltage. When the fluctuating AC component is present in DC current, it is known as the **Current ripple** while the fluctuating AC component in DC voltage is known as the **Voltage ripple**.





QUESTIONS?

SEE YOU IN THE NEXT CLASS

